

Unsteady Aerodynamics in Model-Based Wind Estimation from Fixed-Wing Aircraft Motion

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This paper examines the value of incorporating unsteady aerodynamics in a model-based wind estimator for a small, fixed-wing aircraft and presents results of an experimental implementation. An extended Kalman filter is designed for two motion models identified from flight data: one that includes unsteady effects and another that does not. An air data unit with wind vanes and a Kiel probe is used to reconstruct the local wind velocity for validation. Analysis of time histories of the wind estimates and the wind estimate error relative to the reconstructed wind velocity shows that the two model-based estimators perform equally well. The power spectral densities of the reconstructed wind and the estimated wind, computed using either estimator, follow Kolmogorov's $-5/3$ law.

Nomenclature

$A_{(\cdot)\bar{x}}, B_{(\cdot)\bar{x}}$	=	unsteady aerodynamic parameters, 1/s
b	=	wing span, m
C_l, C_m, C_n	=	coefficient of moment acting on aircraft
C_x, C_y, C_z	=	coefficient of force acting on aircraft
$\bar{c}$	=	mean aerodynamic chord, m
D	=	aircraft propeller diameter, m
$I_{xx}, I_{yy}, I_{zz}, I_{xz}$	=	aircraft moments of inertia, $\text{kg} \cdot \text{m}^2$
m	=	mass of aircraft, kg
S	=	wing surface area, m^2
t	=	time, s
V_r	=	true airspeed, m/s
V_w	=	wind velocity vector expressed in inertial frame, m/s
$\mathbf{v} = [u, v, w]^T$	=	body velocity vector, m/s
$\mathbf{v}_r$	=	air-relative body velocity vector, m/s
$\mathbf{v}_w$	=	wind velocity expressed in body-fixed frame, m/s
$\tilde{\mathbf{v}}$	=	measurement noise vector
$\tilde{\mathbf{w}}$	=	process noise vector
$\mathbf{X}$	=	inertial position (equal to $[x, y, z]^T$), m
$\mathbf{x}$	=	augmented system state $\mathbf{x} = [\mathbf{X}^T, \boldsymbol{\Theta}^T, \mathbf{v}_r^T, \boldsymbol{\omega}^T, \tilde{\mathbf{x}}^T, \mathbf{V}_w^T]^T$
$\tilde{\mathbf{x}}$	=	unsteady state vector $\tilde{\mathbf{x}} = [\tilde{x}_y, \tilde{x}_z, \tilde{x}_l, \tilde{x}_m, \tilde{x}_n]^T$
α, β, β_f	=	angle of attack, side-slip angle, and flank angle, rad
$\delta_a, \delta_e, \delta_r$	=	control surface deflection, rad
δ_{tps}	=	propeller rotation speed, rad/s
n_n	=	number of propellers
η_e	=	propeller efficiency
$\boldsymbol{\Theta} = [\phi, \theta, \psi]^T$	=	aircraft attitude, rad
ρ	=	air density, kg/m^3
$\boldsymbol{\omega} = [p, q, r]^T$	=	body angular velocity vector, rad/s
$\boldsymbol{\omega}_w$	=	effective flow angular rate of wind in body-fixed frame, rad/s
$\boldsymbol{\omega}_r$	=	air-relative body angular velocity vector, rad/s

I. Introduction

SMALL uncrewed aircraft can serve as sensor platforms to measure atmospheric properties in situ [1,2]. Inferring atmospheric wind using aircraft-mounted sensors is a well-established concept [3] in which estimation algorithms combine onboard sensor measurements with a motion model; see, for example, Refs. [4–8]. References [9–13] describe wind estimation where measurements are combined with a kinematic motion model. Others have incorporated full dynamic motion models [14–25].

Wind estimation schemes can be classified into direct and indirect measurements. Direct measurements provide the vehicle's air-relative velocity whose vector difference from the total velocity is the wind velocity [26]. Indirect measurements, on the other hand, use various motion data to infer the wind velocity using a well-designed estimation algorithm [27]. Small, low-cost, uncrewed aircraft are usually equipped with low-cost sensors with a high measurement uncertainty that affects the accuracy of state estimates. Several studies have explored using different sensors for wind estimation to address this issue.

In Ref. [28], only an inertial measurement unit (IMU) was used, while Ref. [29] incorporated both an IMU and magnetometers. In Refs. [25,30], a combination of global navigation satellite system (GNSS) data and barometric pressure sensors improved wind estimate results. In Refs. [12,15,20,22,31–35], various combinations of IMU, magnetometer, GNSS, barometer, and air data unit (ADU) sensors were employed to estimate wind velocity and improve the accuracy of the estimates.

Prior studies have also explored the effect of various estimation algorithms on wind estimate accuracy. Linear state estimators, such as the Kalman filter [14,18,25,36] and the H_∞ filter [37], have been widely used for wind velocity estimation, though they are only expected to be effective for small perturbations from a nominal flight condition. Nonlinear estimators, such as the extended Kalman filter, have also been employed in several studies to combat the shortcomings of linear approaches [19,20,22–24]. The unscented Kalman filter has also been used to infer wind velocity, having the benefit of not requiring computations of Jacobians and better handling of statistical propagation through nonlinearities [15–17,21,38]. Recently, other nonlinear approaches have been used that aim to provide stronger convergence guarantees by leveraging the nonlinear system dynamics [39,40]. Wind velocity estimation error due to low-fidelity dynamic models has received less attention. In Ref. [41], the effect of linearizing a quasi-steady, nonlinear flight dynamic model and ignoring nonuniform flow effects on wind estimation accuracy was explored.

In this paper, the effect of dynamic modeling assumptions on reconstructed wind field states is revisited, except that the model-based estimator (an extended Kalman filter) is based on an unsteady aerodynamic model. We compare the results of wind velocity

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estimation obtained using the unsteady aerodynamic model-based approach with those obtained using a quasi-steady aerodynamic model. Furthermore, we compare both approaches with the reconstructed wind velocity derived from measurements obtained using an ADU. Additionally, we analyze and compare the power spectra of the estimated wind velocity and the reconstructed wind velocity.

In Sec. II, we present background information such as frames of reference, the wind field, and the dynamic equations for aircraft in wind, including aerodynamic force and moment models. In Sec. III, we present the augmented equations of motion, including the extended state variables that represent the wind velocity estimates, the measurement model, and the initialization and tuning parameters for the estimator. In Sec. IV, we describe the test aircraft and its instrumentation, the flight profile flown by the test aircraft, and the reconstruction technique for obtaining the *true* wind velocity from ADU measurements, including analysis of the reconstructed wind velocity. Section V presents the reconstruction of the wind and its analysis. Section VI presents the model-based estimation results. Conclusions are presented in Sec. VII.

II. Background

To support the discussion of wind velocity estimation, we summarize the relevant frames of reference, review some facts and common assumptions about wind velocity fluctuations, and recall the equations of motion for a rigid, fixed-wing aircraft in wind.

A. Reference Frames

In this paper, every reference frame is identified with a right-handed, orthonormal triad of basis vectors. We consider two frames [42–44]:

1) The *north–east–down* (NED) reference frame is represented by an orthonormal triad ($\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$) based at an arbitrary (nonpolar) point on the Earth's surface, with $\mathbf{e}_x$ pointing north, $\mathbf{e}_y$ pointing east, and $\mathbf{e}_z$ pointing down. Over the time and space scales of interest in this paper, the NED frame is assumed to be fixed in inertial space so that Newton's laws of motion hold. We will interchange the terms *NED* and *inertial* in describing this frame. Moreover, we take Earth to be flat, and we assume the force of gravity does not vary with position.

2) The *body-fixed* reference frame is represented by the orthonormal triad ($\mathbf{b}_x, \mathbf{b}_y, \mathbf{b}_z$). The origin of the body-fixed reference frame is at the aircraft's center of gravity, with the positive $\mathbf{b}_x$ axis pointing forward through the nose of the aircraft, the positive $\mathbf{b}_y$ axis pointing out the aircraft's right wing, and the positive $\mathbf{b}_z$ axis pointing down through the belly of the aircraft.

B. Wind Velocity Field

The wind is a three-dimensional vector field that varies in time and space. We denote the wind velocity vector at some position $\mathbf{X} = [x, y, z]^T$ and time t as

$$\mathbf{W}(\mathbf{X}, t) = [W_x(\mathbf{X}, t), W_y(\mathbf{X}, t), W_z(\mathbf{X}, t)]^T \quad (1)$$

The wind velocity vector can be decomposed into a slowly varying, deterministic mean value $\bar{\mathbf{W}}(\mathbf{X}, t)$ and fast-varying, random fluctuations $\tilde{\mathbf{W}}(\mathbf{X}, t)$ from the mean:

$$\mathbf{W}(\mathbf{X}, t) = \bar{\mathbf{W}}(\mathbf{X}, t) + \tilde{\mathbf{W}}(\mathbf{X}, t) \quad (2)$$

The rate of change of the wind velocity as seen by a particle moving along a trajectory $\mathbf{X}(t)$ is

$$\frac{d}{dt} \mathbf{W}(\mathbf{X}, t) = \frac{\partial}{\partial t} \bar{\mathbf{W}} + \nabla \bar{\mathbf{W}} \frac{d\mathbf{X}}{dt} + \frac{\partial}{\partial t} \tilde{\mathbf{W}} + \nabla \tilde{\mathbf{W}} \frac{d\mathbf{X}}{dt} \quad (3)$$

If we assign a body-fixed reference frame to the particle at position $\mathbf{X}(t)$ to represent an aircraft, the spatial gradient of the wind velocity vector can be expressed in this frame of reference as

$$\nabla \mathbf{v}_w = \mathbf{R}_{IB}^T \nabla \mathbf{W} \mathbf{R}_{IB} = \begin{bmatrix} \frac{\partial u_w}{\partial x} & \frac{\partial u_w}{\partial y} & \frac{\partial u_w}{\partial z} \\ \frac{\partial v_w}{\partial x} & \frac{\partial v_w}{\partial y} & \frac{\partial v_w}{\partial z} \\ \frac{\partial w_w}{\partial x} & \frac{\partial w_w}{\partial y} & \frac{\partial w_w}{\partial z} \end{bmatrix} \quad (4)$$

where $\mathbf{v}_w = [u_w, v_w, w_w]^T$ is the wind velocity vector expressed in the body frame and $\mathbf{R}_{IB}$ is the rotation matrix that maps free vectors from the body frame to the inertial frame; the rotation matrix $\mathbf{R}_{IB}$ is discussed in Sec. II.C. The spatial gradient matrix in Eq. (4) can be decomposed into a symmetric part representing dilation and a skew-symmetric part representing rotation. For a fixed-wing aircraft, the effect of dilation can be ignored. The effect of rotation is dominated by certain aspects of the aircraft geometry such that the effective flow angular rate $\boldsymbol{\omega}_w = [p_w, q_w, r_w]^T$ felt by the aircraft is [45]

$$p_w = -\frac{\partial w_w}{\partial y} \quad q_w = \frac{\partial w_w}{\partial x} \quad r_w = \frac{\partial v_w}{\partial x} \quad (5)$$

In this paper, we neglect the gradient of the mean wind velocity $\bar{\mathbf{W}}$ due to the nature of the aircraft's flight. For fixed-wing aircraft flying straight and level at a constant speed and altitude, the aircraft speed is often significantly higher than the mean wind velocity and the velocity fluctuations. As a result, the turbulent fluctuations can be treated as a frozen pattern advecting with the mean wind velocity $\bar{\mathbf{W}}$ (Ref. [43] chap. 13). Summarizing the mentioned points, we assume the following conditions hold over the scale of the aircraft motion.

Assumption 1: The wind velocity is the superposition of a steady, uniform mean component and a spatially varying fluctuation from this mean. More specifically,

- 1) The mean wind is constant in time (steady): $\frac{\partial}{\partial t} \bar{\mathbf{W}}(\mathbf{X}, t) = \mathbf{0}$.
- 2) The mean wind is constant in space (uniform): $\nabla \bar{\mathbf{W}}(\mathbf{X}, t) = \mathbf{0}$.
- 3) The turbulent fluctuations vary only in space (Taylor's hypothesis): $\frac{\partial}{\partial t} \tilde{\mathbf{W}}(\mathbf{X}, t) = \mathbf{0}$.

In practice, the mean wind will vary slowly over the course of a flight. These variations are captured by measurement updates to model-based estimates. In wind estimation, the dynamics of the total wind velocity are often assumed to be driven by a Gaussian white noise process. For the scenario considered here,

$$\frac{d}{dt} \mathbf{W}(\mathbf{X}, t) = \frac{d}{dt} (\bar{\mathbf{W}} + \tilde{\mathbf{W}}(\mathbf{X})) = \nabla \tilde{\mathbf{W}} \frac{d\mathbf{X}}{dt} = \tilde{\mathbf{w}}_w \quad (6)$$

where the stochastic wind velocity fluctuation $\tilde{\mathbf{W}}$ is assumed to be a continuous-time random walk whose generalized derivative $\tilde{\mathbf{w}}_w$ is continuous-time white noise (Ref. [46] chap. 3). Finally, we define the apparent wind velocity vector $\mathbf{V}_w(t)$ as the velocity of the wind expressed in the inertial frame at the current aircraft position $\mathbf{X}(t)$ assuming that the aircraft is absent. That is, the wind velocity vector $\mathbf{V}_w$ is the evaluation of the wind vector field $\mathbf{W}$ along the aircraft's trajectory:

$$\mathbf{V}_w(t) = [V_{w,x}(t), V_{w,y}(t), V_{w,z}(t)]^T := \mathbf{W}(\mathbf{X}(t), t) \quad (7)$$

C. Equations of Motion in Wind

This section presents a flight dynamic model for a rigid aircraft with constant mass subject to aerodynamic forces and moments due to its motion through a wind field. An aircraft's orientation can be described using conventional roll–pitch–yaw Euler angles $\boldsymbol{\Theta} = [\phi, \theta, \psi]^T$. When using this 3–2–1 Euler angle parameterization, the rotation matrix $\mathbf{R}_{IB}$ that maps free vectors from the body-fixed frame to the inertial frame is

$$\mathbf{R}_{IB} = \begin{bmatrix} \cos \theta \cos \psi & \cos \psi \sin \theta \sin \phi - \cos \phi \sin \psi & \cos \psi \sin \theta \cos \phi + \sin \phi \sin \psi \\ \cos \theta \sin \psi & \sin \phi \sin \theta \sin \psi + \cos \phi \cos \psi & \sin \theta \cos \phi \sin \psi - \sin \phi \sin \psi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (8)$$

The aircraft's total translational and angular velocity for motion in wind are (Ref. [47] chaps. 4 and 5)

$$\mathbf{v} = \mathbf{v}_r + \mathbf{R}_{IB}^T \mathbf{V}_w \quad (9a)$$

$$\boldsymbol{\omega} = \boldsymbol{\omega}_r + \boldsymbol{\omega}_w \quad (9b)$$

where $\mathbf{v} = [u, v, w]^T$ is the total aircraft velocity vector expressed in the body-fixed frame, $\mathbf{v}_r = [u_r, v_r, w_r]^T$ is the air-relative aircraft velocity vector expressed in the body-fixed frame, $\boldsymbol{\omega} = [p, q, r]^T$ is the total angular velocity vector, $\boldsymbol{\omega}_r = [p_r, q_r, r_r]^T$ is the air-relative angular velocity vector, and $\boldsymbol{\omega}_w = [p_w, q_w, r_w]^T$ is the effective flow angular rate.

The inertial velocity is $\dot{\mathbf{X}} = \mathbf{R}_{IB} \mathbf{v}_r + \mathbf{V}_w$. The Euler angle rates are $\dot{\boldsymbol{\Theta}} = \mathbf{L}_{IB} \boldsymbol{\omega}$, where

$$\mathbf{L}_{IB} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \quad (10)$$

Using Newton's second law of motion (Ref. [47] chap. 4), the complete equations of motion for a rigid aircraft in wind are

$$\dot{\mathbf{X}} = \mathbf{R}_{IB} \mathbf{v}_r + \mathbf{V}_w \quad (11a)$$

$$\dot{\boldsymbol{\Theta}} = \mathbf{L}_{IB} \boldsymbol{\omega} \quad (11b)$$

$$\dot{\mathbf{v}}_r = \mathbf{v}_r \times \boldsymbol{\omega} + \frac{1}{m} \mathbf{F}_A + \frac{1}{m} \mathbf{R}_{IB}^T \mathbf{F}_G - \mathbf{R}_{IB}^T \dot{\mathbf{V}}_w \quad (11c)$$

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{I}\boldsymbol{\omega} \times \boldsymbol{\omega}) + \mathbf{I}^{-1} \mathbf{M}_A \quad (11d)$$

The vector $\mathbf{F}_A = [F_X, F_Y, F_Z]^T$ is the aerodynamic force vector expressed in the body-fixed frame; $\mathbf{F}_G = [0, 0, mg]^T$ is the weight, which acts down in the inertial frame; and $\mathbf{M}_A = [M_l, M_m, M_n]^T$ is the aerodynamic moment vector expressed in the body-fixed frame and applied about the aircraft's center of gravity. Because of symmetry, the aircraft moment of inertia matrix $\mathbf{I}$ computed in the body-fixed frame is

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix} \quad (12)$$

We consider the control input vector $\boldsymbol{\delta} = [\delta_e, \delta_a, \delta_r, \delta_{\text{tps}}]^T$, which contains the elevator, aileron, and rudder deflections and the propeller rotational speed. In this paper, we model the aerodynamic forces and moments for fixed-wing aircraft using nondimensional force and moment coefficients,

$$\mathbf{F}_A = \frac{1}{2} \rho \|\mathbf{v}_r\|^2 S \begin{bmatrix} C_X \\ C_Y \\ C_Z \end{bmatrix} + D^4 \rho \eta_e \eta_n \delta_{\text{tps}}^2 \begin{bmatrix} C_J \\ 0 \\ 0 \end{bmatrix} \quad (13a)$$

$$\mathbf{M}_A = \frac{1}{2} \rho \|\mathbf{v}_r\|^2 S \begin{bmatrix} b C_l \\ \bar{c} C_m \\ b C_n \end{bmatrix} \quad (13b)$$

where $\bar{c}$ is the mean aerodynamic chord, b is the wingspan, S is the wing planform area, ρ is the air density, η_e is the efficiency parameter

for the aircraft's propellers, η_n is the number of propellers of the aircraft, and D is the diameter of the aircraft's propellers.

D. Aerodynamic Force and Moment Coefficients

In this paper, two aerodynamic models are used: a quasi-steady model and an unsteady model. The equations for the unsteady aerodynamic model are

$$C_J = C_{J_0} + C_J J + C_{J^2} J^2 \quad (14a)$$

$$C_X = C_{X_0} + C_{X_\alpha} \Delta \alpha + C_{X_q} \hat{q} + C_{X_{u_r}} \Delta u_r \quad (14b)$$

$$C_Y = C_{Y_{\tilde{x}}} \tilde{x}_Y + C_{Y_0} + C_{Y_\beta} \Delta \beta + C_{Y_p} \hat{p} + C_{Y_r} \hat{r} + C_{Y_{\delta_a}} \Delta \delta_a + C_{Y_{\delta_r}} \Delta \delta_r \quad (14c)$$

$$C_Z = C_{Z_{\tilde{x}}} \tilde{x}_Z + C_{Z_0} + C_{Z_\alpha} \Delta \alpha + C_{Z_q} \hat{q} + C_{Z_{\delta_e}} \Delta \delta_e \quad (14d)$$

$$C_l = C_{l_{\tilde{x}}} \tilde{x}_l + C_{l_0} + C_{l_\beta} \Delta \beta + C_{l_p} \hat{p} + C_{l_r} \hat{r} + C_{l_{\delta_a}} \Delta \delta_a \quad (14e)$$

$$C_m = C_{m_{\tilde{x}}} \tilde{x}_m + C_{m_0} + C_{m_\alpha} \Delta \alpha + C_{m_q} \hat{q} + C_{m_{\delta_e}} \Delta \delta_e \quad (14f)$$

$$C_n = C_{n_{\tilde{x}}} \tilde{x}_n + C_{n_\beta} \Delta \beta + C_{n_p} \hat{p} + C_{n_r} \hat{r} + C_{n_{\delta_a}} \Delta \delta_a + C_{n_{\delta_r}} \Delta \delta_r \quad (14g)$$

where J is the propeller advance ratio, α is the aircraft's angle of attack, and β is the aircraft's side-slip angle. Note that the model parameters are linear functions of the state variables, with exception of the coefficient C_J , which is quadratic in the advance ratio. The variables that denote perturbations from nominal state and control values given in Eq. (14) are defined as follows:

$$\begin{aligned} \Delta u_r &= u_r - u_{r,\text{nom}} & \Delta \alpha &= \alpha - \alpha_{\text{nom}} & \Delta \beta &= \beta - \beta_{\text{nom}} \\ \Delta \delta_a &= \delta_a - \delta_{a,\text{nom}} & \Delta \delta_e &= \delta_e - \delta_{e,\text{nom}} & \Delta \delta_r &= \delta_r - \delta_{r,\text{nom}} \end{aligned}$$

The nondimensional state variables given in Eq. (14) are defined as follows:

$$\begin{aligned} J &= \frac{u_r}{\delta_{\text{tps}} D} & \alpha &= \tan^{-1} \left(\frac{w_r}{u_r} \right) & \beta &= \sin^{-1} \left(\frac{w_r}{\|\mathbf{v}_r\|} \right) \\ \hat{p} &= \frac{pb}{2V_0} & \hat{q} &= \frac{q\bar{c}}{2V_0} & \hat{r} &= \frac{rb}{2V_0} \end{aligned}$$

The vector $\tilde{\mathbf{x}} = [\tilde{x}_Y, \tilde{x}_Z, \tilde{x}_l, \tilde{x}_m, \tilde{x}_n]^T$ is the nondimensional unsteady aerodynamic state vector with dynamics described by the equations

$$\dot{\tilde{x}}_Y = A_{Y_{\tilde{x}}} \tilde{x}_Y + B_{Y_{\tilde{x}}} \Delta \beta \quad (15a)$$

$$\dot{\tilde{x}}_Z = A_{Z_{\tilde{x}}} \tilde{x}_Z + B_{Z_{\tilde{x}}} \Delta \alpha \quad (15b)$$

$$\dot{\tilde{x}}_l = A_{l_{\tilde{x}}} \tilde{x}_l + B_{l_{\tilde{x}}} \Delta \beta \quad (15c)$$

$$\dot{\tilde{x}}_m = A_{m_{\tilde{x}}} \tilde{x}_m + B_{m_{\tilde{x}}} \Delta \alpha \quad (15d)$$

$$\dot{\tilde{x}}_n = A_{n_{\tilde{x}}} \tilde{x}_n + B_{n_{\tilde{x}}} \Delta \beta \quad (15e)$$

where $A_{(\cdot)_{\tilde{x}}}$ and $B_{(\cdot)_{\tilde{x}}} = 1 \text{ s}^{-1}$ parameterize the unsteady aerodynamics. For compactness, the unsteady dynamic equations given in Eq. (15) are combined as follows:

$$\dot{\tilde{\mathbf{x}}} = \underbrace{\begin{bmatrix} A_{Y_{\tilde{x}}} & 0 & 0 & 0 & 0 \\ 0 & A_{Z_{\tilde{x}}} & 0 & 0 & 0 \\ 0 & 0 & A_{l_{\tilde{x}}} & 0 & 0 \\ 0 & 0 & 0 & A_{m_{\tilde{x}}} & 0 \\ 0 & 0 & 0 & 0 & A_{n_{\tilde{x}}} \end{bmatrix}}_{\tilde{\mathbf{A}}} \tilde{\mathbf{x}} + \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\tilde{\mathbf{B}}} \underbrace{\begin{bmatrix} \Delta\alpha \\ \Delta\beta \\ \ddot{u} \end{bmatrix}}_{\tilde{\mathbf{u}}} \quad (16)$$

The quasi-steady aerodynamic model is structurally identical to Eq. (14), except that the $\tilde{x}$ terms are omitted because the model is quasi-steady. See Appendix B for parameter values. The implementation of an estimation algorithm using the unsteady aerodynamics in Eq. (16) is discussed in Sec. III.A.

III. Wind Velocity Estimation

This section describes the augmented motion equations used as the estimator's propagation model. The measurement model is then described, followed by the design and tuning of the filter.

A. Augmented Equations of Motion with Wind Disturbance

To estimate wind velocity, the state variable V_w is appended to the original state vector to represent the unknown wind velocity in the dynamic model. As discussed in Sec. II.B, the wind dynamics are modeled as a continuous-time random walk in Eq. (6). The wind-augmented system model, with and without unsteady aerodynamic terms, is used as the dynamic propagation model in an extended Kalman filter. The augmented motion model is

$$\dot{\mathbf{X}} = \mathbf{R}_{IB} \mathbf{v}_r + \mathbf{V}_w \quad (17a)$$

$$\dot{\boldsymbol{\Theta}} = \mathbf{L}_{IB} \boldsymbol{\omega} \quad (17b)$$

$$\dot{\mathbf{v}}_r = \mathbf{v}_r \times \boldsymbol{\omega} + \frac{1}{m} \mathbf{F}_A(\mathbf{v}_r, \boldsymbol{\omega}, \delta, \tilde{\mathbf{x}}) + \frac{1}{m} \mathbf{R}_{IB}^T \mathbf{F}_G - \mathbf{R}_{IB}^T \dot{\mathbf{V}}_w \quad (17c)$$

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{I}\boldsymbol{\omega} \times \boldsymbol{\omega}) + \mathbf{I}^{-1} \mathbf{M}_A(\mathbf{v}_r, \boldsymbol{\omega}, \delta, \tilde{\mathbf{x}}) \quad (17d)$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\tilde{\mathbf{u}}(\mathbf{v}_r) \quad (17e)$$

$$\dot{\mathbf{V}}_w = \tilde{\mathbf{w}}_w \quad (17f)$$

In Eq. (17a), $\mathbf{V}_w$ is treated as part of the system state as opposed to an external disturbance. The augmented equations of motion can be written in the compact form,

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}, \delta) + \tilde{\mathbf{w}} \quad (18)$$

where $\mathbf{x} = [\mathbf{X}^T, \boldsymbol{\Theta}^T, \mathbf{v}_r^T, \boldsymbol{\omega}^T, \tilde{\mathbf{x}}^T, \mathbf{V}_w^T]^T$ is the 20×1 augmented state vector and $\tilde{\mathbf{w}} = [\tilde{\mathbf{w}}_X^T, \tilde{\mathbf{w}}_{\Theta}^T, \tilde{\mathbf{w}}_{v_r}^T, \tilde{\mathbf{w}}_{\omega}^T, \tilde{\mathbf{w}}_{\tilde{x}}^T, \tilde{\mathbf{w}}_W^T]^T$ is a corresponding 20×1 disturbance vector of stationary, zero-mean, Gaussian white noise. The 3×1 vectors $\tilde{\mathbf{w}}_X$ and $\tilde{\mathbf{w}}_{\Theta}$ are process noise signals, which help account for error due to the linearization of Eqs. (17a) and (17b). Similarly, the 3×1 process noise vectors $\tilde{\mathbf{w}}_{v_r}$ and $\tilde{\mathbf{w}}_{\omega}$ are intended to account for error due to the linearization of Eqs. (17c) and (17d), as well as aerodynamic modeling error in $\mathbf{F}_A$ and $\mathbf{M}_A$. Additionally, we lump the random process $-\mathbf{R}_{IB}^T \dot{\mathbf{V}}_w$ in Eq. (17c) together with $\tilde{\mathbf{w}}_{v_r}$; see Appendix A for details. Finally, the 5×1 signal $\tilde{\mathbf{w}}_{\tilde{x}}$ accounts for modeling error in the unsteady aerodynamic approximation (17e).

B. Measurement Model

The measurements provided to the estimator are the aircraft position, attitude, inertial velocity (denoted $\mathbf{V}$), and body angular velocity. Note that the 12×1 measurement vector $\mathbf{z}$ and the augmented state vector $\mathbf{x}$ have a nonlinear relationship because of the attitude dependent wind triangle expression (9a),

$$\mathbf{z}(t) = \mathbf{h}(\mathbf{x}) + \tilde{\mathbf{v}}(t) = \begin{bmatrix} \mathbf{X} \\ \boldsymbol{\Theta} \\ \mathbf{R}_{IB} \mathbf{v}_r + \mathbf{V}_w \\ \boldsymbol{\omega} \end{bmatrix} + \tilde{\mathbf{v}} = \begin{bmatrix} \mathbf{X} \\ \boldsymbol{\Theta} \\ \mathbf{V} \\ \boldsymbol{\omega} \end{bmatrix} + \tilde{\mathbf{v}} \quad (19)$$

where $\tilde{\mathbf{v}}$ is the 12×1 measurement noise vector assumed to be zero-mean, Gaussian, white noise. Local observability of the system (18) with measurements (19) was verified along the aircraft's nominal motion – steady, wings level flight at constant speed and altitude – for both the quasi-steady and unsteady system.

C. Continuous–Discrete Extended Kalman Filter

Because the experimental aircraft is a sampled data system, we adopt the hybrid (continuous–discrete) extended Kalman filter; the algorithm is briefly provided in Appendix C. The system dynamics (18) are continuous time, but the measurement model is discrete time:



Fig. 1 My Twin Dream aircraft.

Table 1 My Twin Dream aircraft properties

Property	Symbol	Value
Mass, kg	m	3.311
Mean aerodynamic chord, m	$\bar{c}$	0.254
Wing span, m	b	1.80
Wing planform area, m ²	S	0.457
Number of propellers	η_n	2
Propeller efficiency	η_e	0.9
Propeller diameter, m	D	0.254
Roll moment of inertia, kg · m ²	I_{xx}	0.319
Pitch moment of inertia, kg · m ²	I_{yy}	0.267
Yaw moment of inertia, kg · m ²	I_{zz}	0.471
Product of inertia, kg · m ²	I_{xz}	0.024

Table 2 Autopilot sensor suite

Sensor package	Sensors
TDK InvenSense ICM20948	Accelerometer Gyroscope Magnetometer
TDK InvenSense ICM20649	Accelerometer Gyroscope
TDK InvenSense ICM20602	Accelerometer Gyroscope
MEAS Switzerland MS5611	Barometer
u-blox NEO-M8P	GNSS

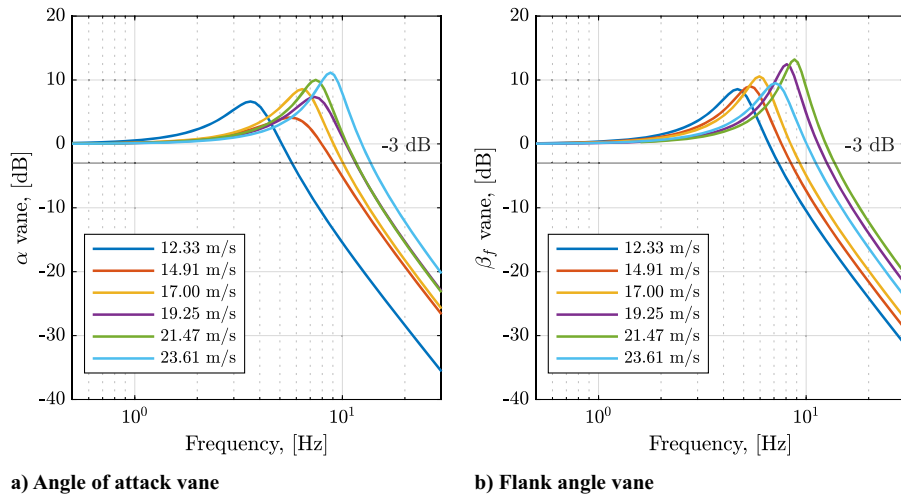


Fig. 2 Frequency response of a) angle of attack and b) flank angle vanes.

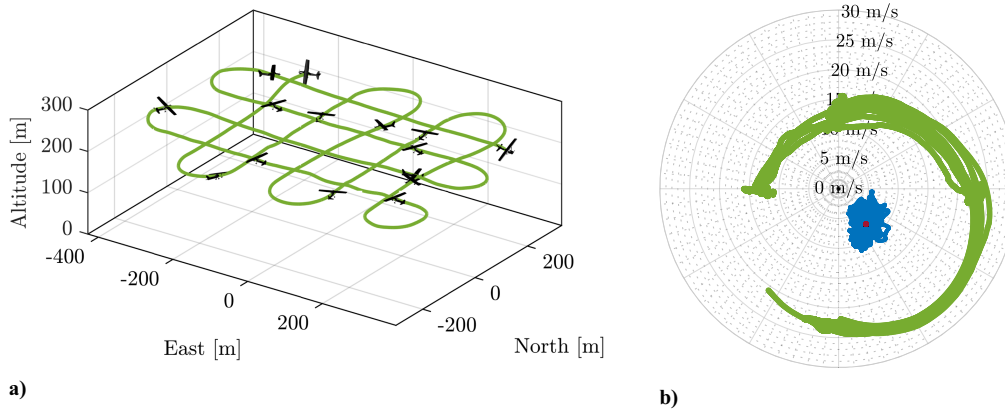


Fig. 3 Flight path (left) and aircraft and wind velocity (right): a) experimental flight profile and b) the green dots indicating the aircraft's horizontal speed, the blue dots indicating the horizontal wind speed and direction, and the red dot indicating the wind's mean velocity and direction.

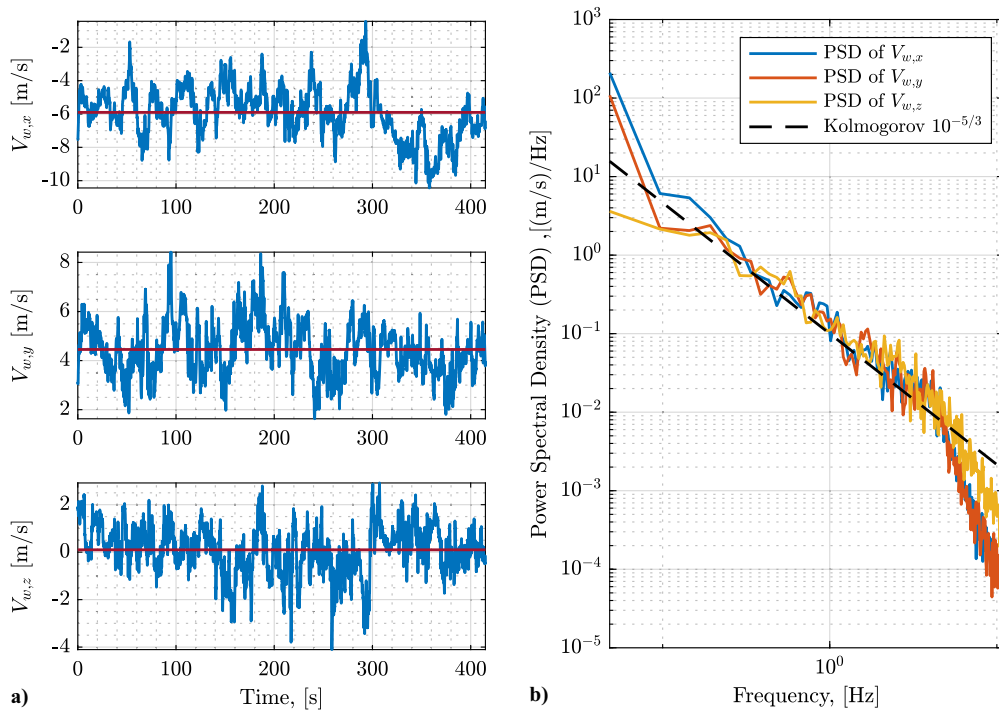


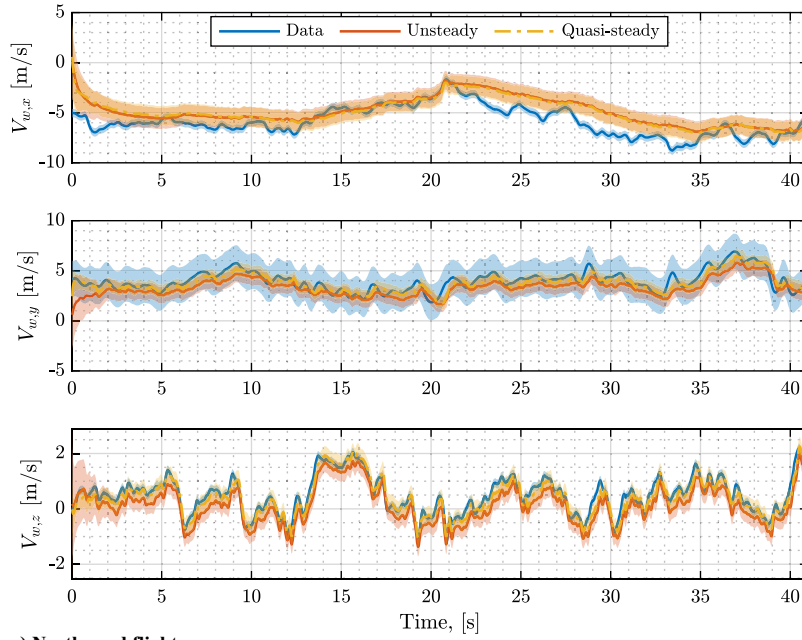
Fig. 4 Reconstructed wind velocity: a) wind velocity components reconstructed from ADU measurements and b) power spectra of the reconstructed wind velocity components.

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \delta(t)) + \tilde{\mathbf{w}}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \quad (20a)$$

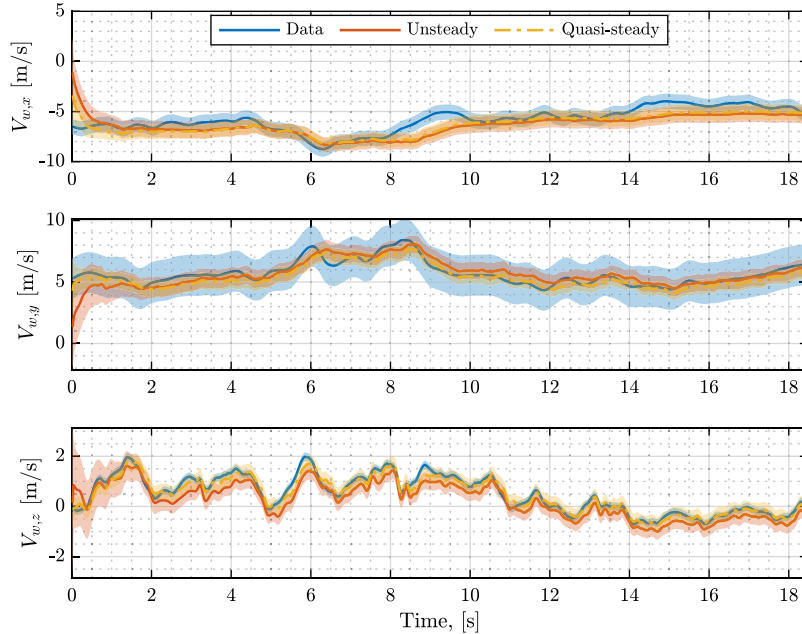
$$\mathbf{z}(t_k) = \mathbf{h}(t_k, \mathbf{x}(t_k)) + \tilde{\mathbf{v}}(t_k), \quad t_k \in \{t_0, t_1, t_2, \dots\} \quad (20b)$$

For a constant sampling period Δt , one has $t_k = t_0 + k\Delta t$; however, the algorithm does not require a fixed sample rate. The hybrid extended Kalman filter's algorithm uses a numerical integration scheme to propagate the state estimate and its covariance from one sample time to the next. The process and measurement noise are assumed to be described by Gaussian distributions with $\tilde{\mathbf{w}}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ and $\tilde{\mathbf{v}}(t_k) \sim \mathcal{N}(\mathbf{0}, \mathbf{R}(t_k))$, respectively. The measurements $\mathbf{z}(t_k)$ are obtained from the flight computer's low-level navigation solution, except for angular velocity, which is obtained directly from the vehicle's angular rate gyroscopes. The associated measurement noise covariance $\mathbf{R}(t_k)$ is taken to be diagonal with entries corresponding to the state estimate variances reported by the flight computer. For the gyroscope readings, the measurement noise covariance

is considered constant and is estimated by smoothing historical data. The initial state estimate $\hat{\mathbf{x}}(t_0)$ and the initial state estimate error covariance $\mathbf{P}(t_0)$ are similarly chosen based on the flight computer's navigation solution. Specifically, the initial estimates and error covariance of position $\hat{\mathbf{X}}(t_0)$ and attitude $\hat{\boldsymbol{\Theta}}(t_0)$ are taken from the flight computer navigation solution. The initial estimate of angular velocity $\hat{\boldsymbol{\omega}}(t_0)$ is taken from the vehicle's gyroscope with a covariance equal to that of its measurement noise. The relative velocity initial estimate $\hat{\mathbf{v}}_r(t_0)$ is computed using the rotation matrix constructed from $\hat{\boldsymbol{\Theta}}(t_0)$ along with the measured inertial velocity $\mathbf{V}$ from the autopilot. Finally, the initial estimates of wind velocity $\hat{\mathbf{V}}_w(t_0)$ and of the unsteady states $\hat{\mathbf{x}}(t_0)$ were taken to be zero with covariances appropriate for their typical magnitudes. The constant process noise power spectral density $\mathbf{Q}$ is obtained by computing the power spectral density of the residuals of the true and modeled state derivative from historical flight data. The residuals are computed as follows:



a) Northward flight



b) Southward flight

Fig. 5 Reconstructed and estimated wind component time histories for north and south flight.

$$\nu = \dot{\mathbf{x}}_{\text{truth}} - \mathbf{f}(t, \mathbf{x}_{\text{truth}}, \boldsymbol{\delta}_{\text{truth}}) \quad (21)$$

The constant power spectral density of the residual is computed using the MATLAB® function `pwelch` as

$$\mathcal{Q} = \max(\text{pwelch}(\nu, [], [], [], f_s)) \quad (22)$$

where f_s is the sampling frequency. Because their true values are unknown, the unsteady states in $\mathbf{x}_{\text{truth}}$ are set to 0 when computing $\mathcal{Q}$.

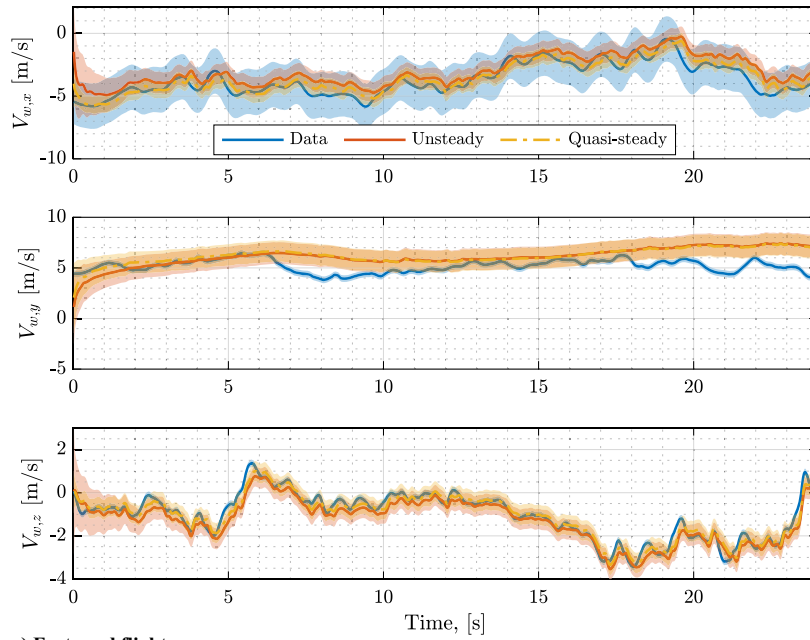
IV. Flight Test

This section describes the research aircraft and its instrumentation as well as the flight test in which wind data were collected to compare the two model-based wind estimators.

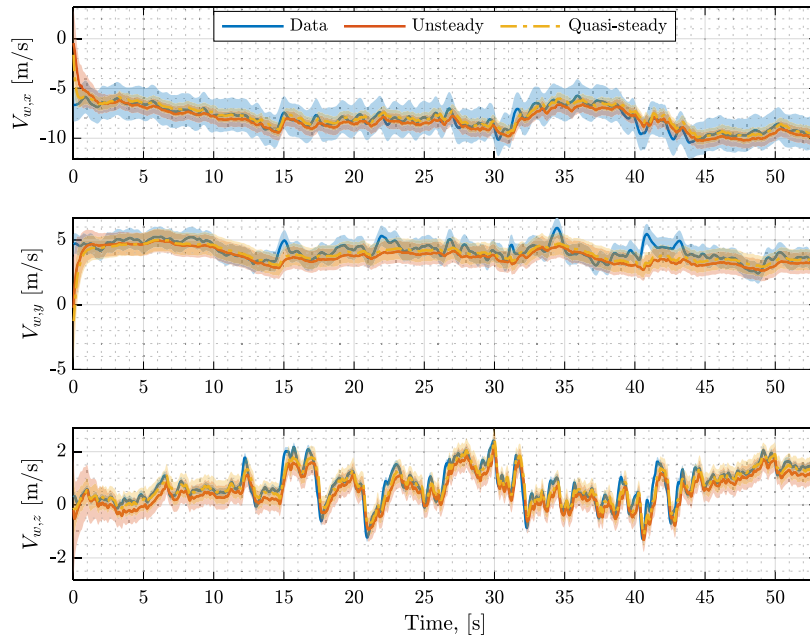
A. Research Aircraft

The research aircraft used, the My Twin Dream (MTD) manufactured by My Fly Dream, is a foam-structured fixed-wing aircraft with a counterrotating twin propeller configuration; see Fig. 1. The aircraft is powered by a four-cell 14.8 V, 6750 mAh lithium-polymer battery. The MTD was chosen because of its simple construction, benign flight dynamics, endurance greater than 20 min, and ample payload capacity for the desired instrumentation package. Mass and geometric properties are shown in Table 1. Sensor information is given in Table 2. Aerodynamic model parameters are shown in Tables B1 and B2 in the Appendix.

The MTD aircraft is instrumented with a Cubepilot Cube Orange flight computer running PX4 firmware for data collection, specifically for its inertial navigation solution (position, attitude, velocity, and angular velocity) [48]. The flight computer sensor suite includes several sensor packages that provide three-component linear acceleration and angular velocity, two magnetometers, a barometer, and a GNSS receiver



a) Eastward flight



b) Westward flight

Fig. 6 Reconstructed and estimated wind component time histories for east and west flight.

capable of real-time kinematic positioning; see Table 2. The pulse-width modulated signals sent from the flight computer to the control surface servos are also recorded. The control surface deflection angle is calculated postflight from a servo-actuator model developed in Refs. [49,50].

The MTD also carries a vaned ADU for measuring air-relative velocity. The ADU, visible in Fig. 1, gives the aircraft's true airspeed V_r , angle of attack α , and flank angle β_f . The dynamic response of the angle of attack and flank angle vanes at different airspeeds can be inferred from Fig. 2, which was obtained from data collected in an open-jet wind tunnel. The process involved applying impulse perturbations to the vane angles at different freestream velocities and then recording the damped, harmonic subsidence from these perturbations. The logarithmic decrement method was used to determine the vanes' damping ratios and natural frequencies at different velocities. The bandwidth of the α and β_f vanes is approximately 6 Hz for a freestream velocity of 12.33 m/s, which is approximately the stall speed of the MTD. The sensor bandwidth increases up to about 14 Hz for a freestream velocity of 23.61 m/s. The cutoff frequency for the ADU vanes is at least two times higher than the MTD's short-period mode natural frequency of about 3 Hz at the nominal cruise speed of 18 m/s. Aerodynamic angle measurements are used only to validate the wind estimates generated using other measurement data.

B. Experimental Flight Profile

The flight test in which wind data were collected to support comparative analysis of the two model-based wind estimators was conducted at Virginia Tech Kentland Experimental Aerial Systems Laboratory. The aircraft was flown at an altitude of 210 m above ground level with a cruise speed of 18 m/s in the grid pattern shown in Fig. 3a. The mission was automated, except for takeoff and landing. Figure 3b shows the measured horizontal wind velocity, where the radial distance from the origin to a data point indicates wind speed, and the azimuth angle indicates wind direction in degrees from true north. The blue dots represent individual wind measurements; the red dot indicates the mean wind velocity. The wind was blowing from the northwest to the southeast throughout the flight, with a mean wind speed of 7.55 m/s, varying between 3 and 12 m/s. See Sec. V.A for more details on the reconstruction of wind velocity from flight data.

V. Reconstructed Wind Velocity

In this section, the reconstructed wind obtained using ADU measurements is presented and analyzed.

A. Wind Reconstruction from ADU Measurements

The ADU measurements were collected to reconstruct the wind velocity following the approach of Ref. [51]. The measured true velocity ($V_r = \|\mathbf{v}_r\|$), flank angle β_f , and angle of attack α were used to reconstruct the three-dimensional body-fixed wind velocities as follows:

$$\mathbf{V}_w = \mathbf{V} - \mathbf{R}_{IB}(\boldsymbol{\Theta})(\mathbf{v}_{ADU} - \boldsymbol{\omega} \times \mathbf{r}_{ADU}) \quad (23)$$

Here, $\mathbf{v}_{ADU} = \mathbf{R}_{BW}(\alpha, \beta)[V_r, 0, 0]^T$ is the air-relative velocity at the geometric center of the ADU vanes, whose position in the body frame is denoted $\mathbf{r}_{ADU}$. The rotation matrix

$$\mathbf{R}_{BW} = \begin{bmatrix} \cos(\alpha) \cos(\beta) & -\cos(\alpha) \sin(\beta) & -\sin(\alpha) \\ \sin(\beta) & \cos(\beta) & 0 \\ \sin(\alpha) \cos(\beta) & -\sin(\alpha) \sin(\beta) & \cos(\alpha) \end{bmatrix}$$

maps free vectors from the wind frame to the body frame. Note that the side-slip angle can be computed from the flank angle as $\beta = \tan^{-1}(\tan(\beta_f) \cos(\alpha))$. The aircraft attitude $\boldsymbol{\Theta}$ and angular velocity $\boldsymbol{\omega}$ are acquired from the aircraft's flight computer navigation solution and rate gyroscopes, respectively. Figure 4 concerns the reconstructed wind velocity over the entire flight depicted in Fig. 3. The reconstructed inertial wind velocity components (blue) and their time averages (red) are shown in Fig. 4a. The power spectra (solid) and the Kolmogorov $10^{-5/3}$ reference (dashed) are shown in Fig. 4b.

B. Reconstructed Wind Results and Analysis

Wind data were collected in a grid pattern flight at constant altitude as shown in Fig. 3a. Assumptions (1a) and (1b) concerning the temporal and spatial derivatives of the mean wind $\bar{\mathbf{W}}$ are satisfied by construction because the mean wind is computed as a time average of the full data set. All variation from the mean wind is attributed to turbulent fluctuations.

Assumption (1c), which states that the wind velocity fluctuations $\tilde{\mathbf{W}}$ vary in space but not in time, relates to the relative time scales of aircraft and atmospheric motion. The frozen turbulence assumption is appropriate, for example, for a fixed-wing aircraft flying straight and level, at a constant altitude, and at a speed much higher than the wind velocity. Figure 3b shows the aircraft's speed and the measured wind velocity throughout the flight that is depicted in Fig. 3a. The nominal

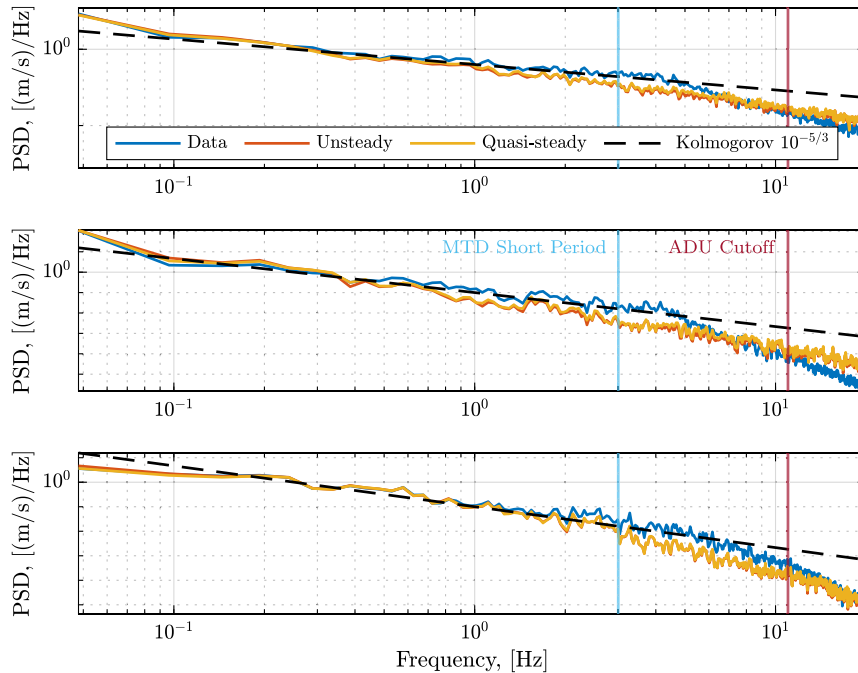


Fig. 7 Reconstructed and estimated wind component power spectral density (PSD) over the entire experimental flight profile.

speed of the aircraft is about four times larger than the measured wind velocity, supporting Assumption (1c) that $\mathbf{W}$ is a *frozen* vector field advecting with the mean wind.

VI. Results and Discussion

This section compares model-based estimates of wind velocity and power spectra using unsteady and quasi-steady aerodynamic models. As shown in Fig. 3a, the aircraft is flown in a grid pattern. Only four legs of the flight are used in the analysis described here, corresponding to northward, southward, eastward, and westward flight. The reconstructed wind velocity and power spectra are taken as true values to which the unsteady and quasi-steady model-based wind velocity and power spectrum estimates are compared.

In the proceeding plots, true values are shown in blue, the quasi-steady aerodynamic model-based wind estimates are shown in gold, and the unsteady aerodynamic model-based wind estimates are shown in burnt orange. The dashed black reference lines in the power

spectrum plots have a slope of $10^{-5/3}$. For the time histories of wind estimates shown in Figs. 5a, 5b, 6a, and 6b, shaded regions indicating ± 1 standard deviation bounds are shown in the respective colors. These standard deviations were computed numerically by propagating samples of normally distributed uncertain values through Eq. (23). Specifically, the standard deviations of Θ , $\mathbf{V}$, and ω were chosen in the same way as explained in Sec. III.C, and the standard deviations of the vane angles and differential pressure sensor-based airspeed measurements were obtained from manufacturer specifications.

In Figs. 5a–6b, the estimated wind velocities, given in gold and burnt orange, are compared to the reconstructed wind velocity given in blue. These values are established from the northward, southward, eastward, and westward legs. The unsteady aerodynamic and quasi-steady aerodynamic model-based wind estimates track the reconstructed wind velocity comparably well.

Figure 7 shows a comparison between the power spectral density of the estimated wind velocities and the reconstructed wind velocity. A large data set is needed to visualize the energy content of wind

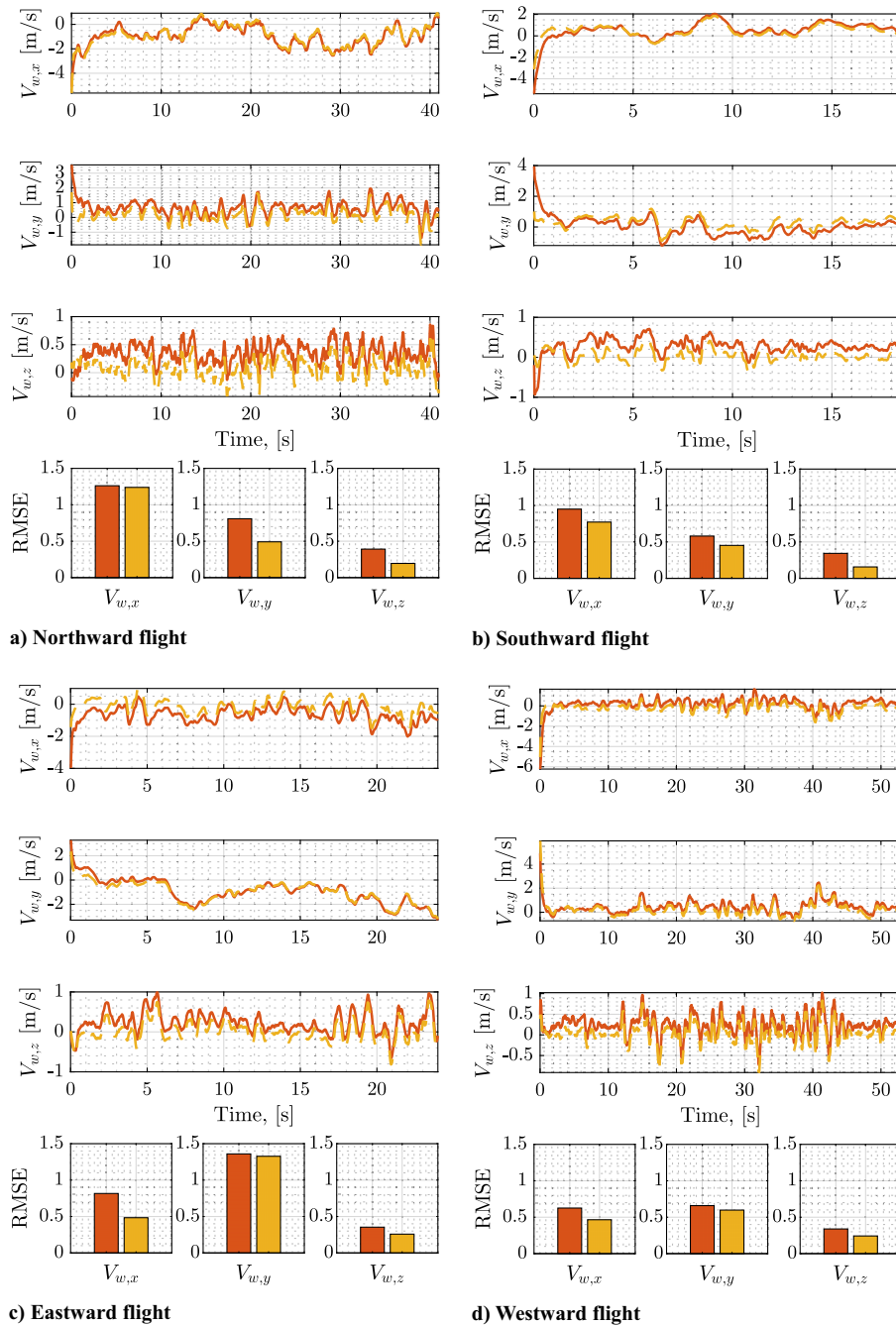


Fig. 8 Time histories of error and root-mean-square error for the estimated wind components. (Burnt orange is unsteady; gold is quasi-steady).

velocities. Hence, the entire flight experiment profile was used to generate the plot in this case. The power spectral density plots for the wind estimates based on both aerodynamic models track the power spectral density of the reconstructed wind. The slope of both the reconstructed and estimated power spectral densities track the dashed black lines with a slope of $10^{-5/3}$, reflecting physically reasonable wind characteristics. Vertical lines in Fig. 7 indicate the aircraft short period mode (around 3 Hz) and the ADU vane cutoff frequency (around 11 Hz). The spectral properties computed for the quasi-steady and unsteady model-based wind estimates are considered most accurate below 3 Hz. Conversely, the spectral properties of the reconstructed wind velocity values are considered unreliable above 11 Hz.

The error plots given in Fig. 8 indicate that the error of both aerodynamic model-based wind estimates relative to the reconstructed wind velocity are low. The bar graphs showing root-mean-square error suggest that the unsteady and the quasi-steady aerodynamic model-based wind estimates are comparable, but with the quasi-steady results consistently showing slightly lower error.

The direction of flight with respect to the direction of the wind affected the performance of the filters. During the field experiment, the measured wind was blowing from the northwest to the southeast with a mean wind velocity of 7.55 m/s, gusting between 3 and 12 m/s. The $V_{w,x}$ estimates improved when the aircraft flew from west to east and from east to west. Similarly, the $V_{w,y}$ estimates improved when the aircraft flew from north to south and from south to north. This suggests the wind component perpendicular to the direction of aircraft motion is estimated more accurately. The (horizontal) direction of aircraft motion did not affect the $V_{w,z}$ wind estimates.

VII. Conclusions

The performance of two model-based wind estimators for a small, fixed-wing aircraft were compared. In one estimator, the flight dynamic model incorporated unsteady aerodynamics identified from flight-test data, while the other estimator used a quasi-steady flight dynamic model. To provide a baseline for this comparison, the wind velocity was reconstructed from measurements obtained using a vaned air data unit. The measurements from this sensor were not incorporated into the model-based wind estimates. The unsteady and quasi-steady aerodynamic model-based wind velocity estimates performed similarly in comparison with the reconstructed wind velocity, with the quasi-steady model-based estimates showing marginally lower root-mean-square error. The power spectral density of both model-based wind estimates agree with the power spectral density of the reconstructed wind velocity, and all three power spectra exhibit Kolmogorov's $10^{-5/3}$ slope, consistent with expectations. It has been shown how unsteady aerodynamic effects may be incorporated into a model-based wind estimation scheme for a fixed-wing aircraft. For the aircraft and flight data considered here, the procedure did not improve wind estimate accuracy, compared with results obtained using a quasi-steady aerodynamic model. It is possible, though, that incorporating unsteady effects could improve model-based wind estimates for other vehicle configurations or environmental conditions.

Appendix: Motion Model and Filter Details

A. Flight Dynamic Model Derivation

Equation (11d) is derived from Newton's second law of motion, written in the noninertial body-fixed frame, as follows. Noting that

$$m \frac{d}{dt} (\mathbf{v}_r + \mathbf{R}_{IB}^T \mathbf{V}_w) = m (\mathbf{v}_r + \mathbf{R}_{IB}^T \mathbf{V}_w) \times \boldsymbol{\omega} + \mathbf{F}_A(\mathbf{v}_r, \boldsymbol{\omega}_r, \boldsymbol{\delta}) + \mathbf{R}_{IB}^T \mathbf{F}_G$$

we compute

$$m (\dot{\mathbf{v}}_r + \dot{\mathbf{R}}_{IB}^T \mathbf{V}_w + \mathbf{R}_{IB}^T \dot{\mathbf{V}}_w) = m \mathbf{v}_r \times \boldsymbol{\omega} + m \mathbf{R}_{IB}^T \mathbf{V}_w \times \boldsymbol{\omega} + \mathbf{F}_A(\mathbf{v}_r, \boldsymbol{\omega}_r, \boldsymbol{\delta}) + \mathbf{R}_{IB}^T \mathbf{F}_G$$

Moving the wind-related terms from the left to the right gives

$$m \dot{\mathbf{v}}_r = m \mathbf{v}_r \times \boldsymbol{\omega} + m \mathbf{v}_w \times \boldsymbol{\omega} + \mathbf{F}_A(\mathbf{v}_r, \boldsymbol{\omega}_r, \boldsymbol{\delta}) + \mathbf{R}_{IB}^T \mathbf{F}_G - m \mathbf{v}_w \times \boldsymbol{\omega} - m \mathbf{R}_{IB}^T \dot{\mathbf{V}}_w$$

Table B1 MTD aerodynamic force parameters

Unsteady model		Quasi-steady model	
Term	$\hat{\theta} \pm \sigma$	Term	$\hat{\theta} \pm \sigma$
C_{X_0}	-0.4048 ± 0.0001	C_{X_0}	-0.4045 ± 0.0002
C_{X_α}	$+0.2610 \pm 0.0040$	C_{X_α}	$+0.2625 \pm 0.0061$
C_{X_q}	$+2.067 \pm 0.0432$	C_{X_q}	$+2.0856 \pm 0.0674$
$C_{X_{\dot{\alpha}}}$	-0.1923 ± 0.0007	$C_{X_{\dot{\alpha}}}$	-0.1935 ± 0.0011
C_{J_0}	$+0.1159 \pm 0.0002$	C_{J_0}	$+0.1159 \pm 0.0002$
C_J	-0.0400 ± 0.0013	C_J	-0.0400 ± 0.0013
C_{J^2}	-0.1310 ± 0.0015	C_{J^2}	-0.1310 ± 0.0015
$A_{Y_{\dot{\alpha}}}$	-1.2739 ± 0.0424	—	—
$C_{Y_{\dot{\alpha}}}$	$+0.1238 \pm 0.0076$	—	—
C_{Y_0}	$+0.0007 \pm 0.0001$	C_{Y_0}	$+0.0004 \pm 0.0002$
C_{Y_β}	-0.5585 ± 0.0014	C_{Y_β}	-0.5524 ± 0.0019
C_{Y_p}	$+0.1928 \pm 0.0041$	C_{Y_p}	$+0.1936 \pm 0.0065$
C_{Y_r}	$+0.4311 \pm 0.0052$	C_{Y_r}	$+0.4731 \pm 0.0069$
$C_{Y_{\delta_a}}$	-0.1052 ± 0.0014	$C_{Y_{\delta_a}}$	-0.1050 ± 0.0022
$C_{Y_{\delta_r}}$	$+0.1644 \pm 0.0012$	$C_{Y_{\delta_r}}$	$+0.1637 \pm 0.0018$
$A_{Z_{\dot{\alpha}}}$	-0.3524 ± 0.0228	—	—
$C_{Z_{\dot{\alpha}}}$	$+0.4179 \pm 0.0375$	—	—
C_{Z_0}	-0.3690 ± 0.0010	C_{Z_0}	-0.3642 ± 0.0015
C_{Z_α}	-5.7611 ± 0.0340	C_{Z_α}	-5.6419 ± 0.0507
C_{Z_q}	-42.712 ± 0.4680	C_{Z_q}	-42.975 ± 0.7844
$C_{Z_{\dot{\alpha}}}$	-0.4581 ± 0.0115	$C_{Z_{\dot{\alpha}}}$	-0.4515 ± 0.0180

$\hat{\theta}$ represents the given parameter estimate, σ represents one standard deviation.

Table B2 MTD aerodynamic moment parameters

Unsteady model		Quasi-steady model	
Term	$\hat{\theta} \pm \sigma$	Term	$\hat{\theta} \pm \sigma$
$A_{l_{\dot{\alpha}}}$	-0.9815 ± 0.0444	—	—
$C_{l_{\dot{\alpha}}}$	-0.0052 ± 0.0023	—	—
C_{l_0}	-0.0006 ± 0.0001	C_{l_0}	-0.0006 ± 0.0001
C_{l_p}	-0.0314 ± 0.0005	C_{l_p}	-0.0316 ± 0.0006
C_{l_r}	-0.2882 ± 0.0014	C_{l_r}	-0.2881 ± 0.0022
$C_{l_{\delta_a}}$	$+0.0684 \pm 0.0016$	$C_{l_{\delta_a}}$	$+0.0669 \pm 0.0021$
$C_{l_{\delta_r}}$	$+0.1283 \pm 0.0005$	$C_{l_{\delta_r}}$	$+0.1281 \pm 0.0008$
$A_{m_{\dot{\alpha}}}$	-1.0186 ± 0.0426	—	—
$C_{m_{\dot{\alpha}}}$	$+0.1908 \pm 0.0142$	—	—
C_{m_0}	$+0.0032 \pm 0.0002$	C_{m_0}	$+0.0039 \pm 0.0002$
C_{m_α}	-0.6273 ± 0.0056	C_{m_α}	-0.5930 ± 0.0077
C_{m_q}	-5.6897 ± 0.0725	C_{m_q}	-5.7774 ± 0.1202
$C_{m_{\dot{\alpha}}}$	-0.354 ± 0.0018	$C_{m_{\dot{\alpha}}}$	-0.3525 ± 0.0028
$A_{n_{\dot{\alpha}}}$	-1.4365 ± 0.0381	—	—
$C_{n_{\dot{\alpha}}}$	-0.0759 ± 0.0024	—	—
C_{n_p}	$+0.0947 \pm 0.0004$	C_{n_p}	$+0.0904 \pm 0.0005$
C_{n_r}	-0.0323 ± 0.0011	C_{n_r}	-0.0328 ± 0.0018
$C_{n_{\delta_a}}$	-0.0871 ± 0.0015	$C_{n_{\delta_a}}$	-0.1118 ± 0.0019
$C_{n_{\delta_r}}$	-0.0011 ± 0.0004	$C_{n_{\delta_r}}$	-0.0014 ± 0.0006
$C_{n_{\dot{\alpha}}}$	-0.0587 ± 0.0003	$C_{n_{\dot{\alpha}}}$	-0.0584 ± 0.0005

$\hat{\theta}$ represents the given parameter estimate, σ represents one standard deviation.

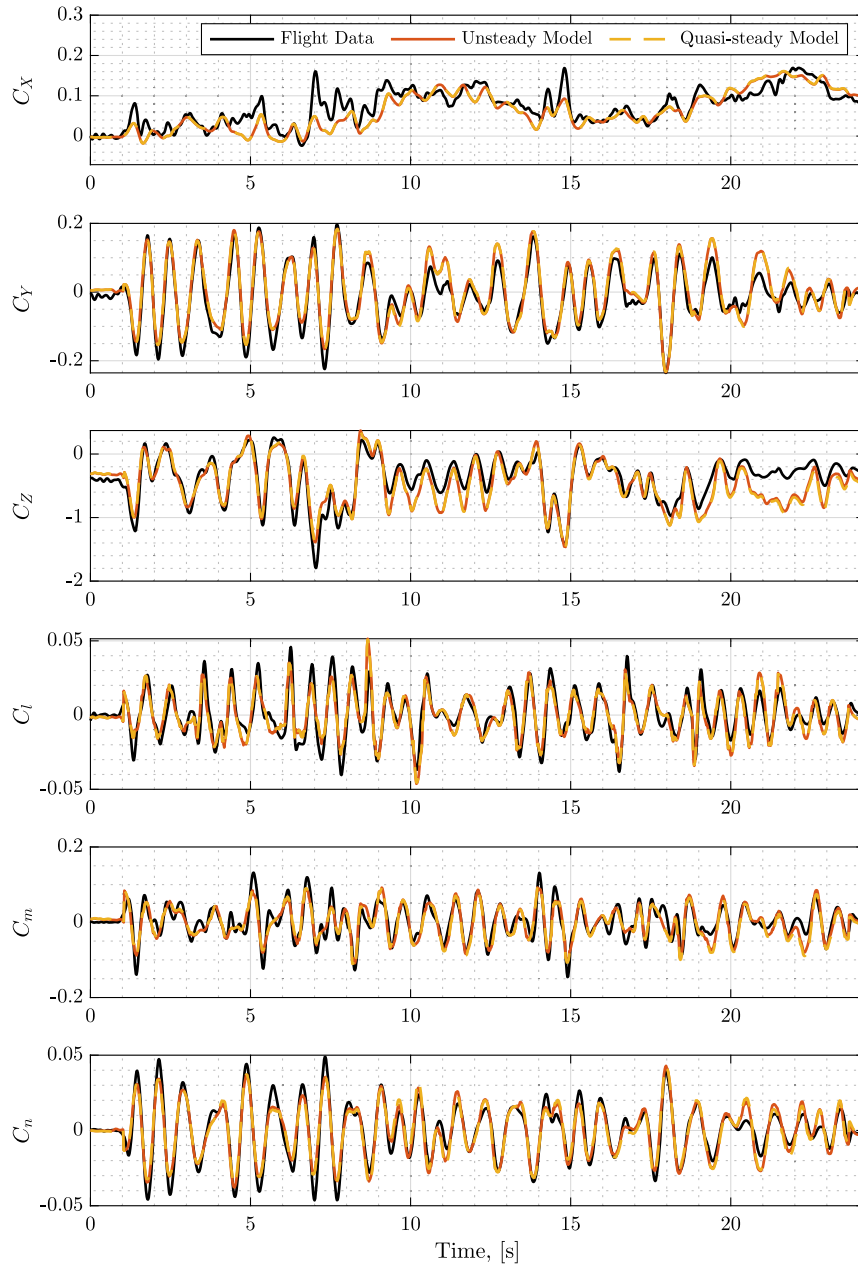


Fig. A1 Quasi-steady and unsteady aerodynamic model validation.

Simplifying, we find

$$m\dot{\mathbf{v}}_r = m\mathbf{v}_r \times \boldsymbol{\omega} + \mathbf{F}_A(\mathbf{v}_r, \boldsymbol{\omega}_r, \boldsymbol{\delta}) + \mathbf{R}_{IB}^T \mathbf{F}_G - m\mathbf{R}_{IB}^T \dot{\mathbf{V}}_w$$

Finally, dividing through by mass gives

$$\dot{\mathbf{v}}_r = \mathbf{v}_r \times \boldsymbol{\omega} + \frac{1}{m} \mathbf{F}_A(\mathbf{v}_r, \boldsymbol{\omega}_r, \boldsymbol{\delta}) + \frac{1}{m} \mathbf{R}_{IB}^T \mathbf{F}_G - \mathbf{R}_{IB}^T \dot{\mathbf{V}}_w$$

In this paper, the last term ($-\mathbf{R}_{IB}^T \dot{\mathbf{V}}_w$) is modeled as noise and is accounted for in the $\tilde{\mathbf{w}}_{v_r}$ term as discussed in Sec. III.A.

B. Aircraft Aerodynamic Parameters

Tables B1 and B2 contain the aerodynamic force and moment parameters introduced in Sec. II.D for the MTD. The identified model is assumed to be valid for the flight domain indicated by Table B3. These explanatory variable upper and lower bounds were taken to be the minimum and maximum values, respectively, in the

training data set after outlier removal using MATLAB® command `rmoutliers`.[§] The model validation results are presented in Fig. A1, and their normalized root-mean-square error (NRMSE) is provided in Table B4. The NRMSE used in this paper is calculated as

$$\text{NRMSE} = \frac{1}{\max(\mathbf{z}_{\text{flight}}) - \min(\mathbf{z}_{\text{flight}})} \sqrt{\frac{\|\mathbf{z}_{\text{flight}} - \hat{\mathbf{y}}_{\text{model}}\|^2}{N}} \quad (24)$$

where $\mathbf{z}_{\text{flight}}$ is the measured output from flight data, $\hat{\mathbf{y}}_{\text{model}}$ is the response of the model, and N is the number of data points in $\mathbf{z}_{\text{flight}}$ and $\hat{\mathbf{y}}_{\text{model}}$. The time histories and NRMSE values for the validation data reflect nearly identical performance for the two models.

[§]Data more than three scaled median absolute deviations from the median are removed. See <https://www.mathworks.com/help/matlab/ref/rmoutliers.html> (retrieved 25 January 2024) for more information.

Table B3 MTD aerodynamic model domain

	Minimum	Mean	Maximum
u_t , m/s	+12.504	+21.096	+30.527
α , rad	-0.0861	+0.0412	+0.1735
β , rad	-0.2626	+0.0524	+0.3672
p , rad/s	-3.1770	+0.0045	+3.1405
q , rad/s	-1.5367	+0.1143	+1.6566
r , rad/s	-1.9676	-0.0252	+1.9204
δ_a , rad	-0.4745	+0.0114	+0.4854
δ_e , rad	-0.4657	-0.0105	+0.4486
δ_r , rad	-0.3013	+0.0081	+0.3306
J	+0.2064	+0.3423	+0.4368

Table B4 Normalized root mean square error

	Unsteady Aerodynamic model	Quasi-steady Aerodynamic model
C_X	+0.3129	+0.3163
C_Y	+0.0624	+0.0636
C_Z	+0.1093	+0.1165
C_l	+0.0744	+0.0745
C_m	+0.0853	+0.0874
C_n	+0.0631	+0.0703

C. Continuous–Discrete (Hybrid) Extended Kalman Filter Algorithm

The estimator used for the estimation problem described in Sec. III is the continuous–discrete (hybrid) extended Kalman filter, which is summarized in Algorithm C1.

Algorithm C1 Continuous–discrete (hybrid) extended Kalman filter

Step 1: Initialization
Start with $\hat{\mathbf{x}}(t_0)$ and $\mathbf{P}(t_0)$
Set $t_k = 0$
Step 2: Dynamic propagation from sample t_k to sample t_{k+1}
$\tilde{\mathbf{x}}(t_{k+1}) = \hat{\mathbf{x}}(t_k) + \int_{t_k}^{t_{k+1}} \mathbf{f}(\boldsymbol{\tau}, \hat{\mathbf{x}}(\boldsymbol{\tau}), \boldsymbol{\delta}(\boldsymbol{\tau})) d\boldsymbol{\tau}$
$\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \Big _{[\mathbf{t}, \hat{\mathbf{x}}(\mathbf{t}), \boldsymbol{\delta}(\mathbf{t})]}$
$\tilde{\mathbf{P}}(t_{k+1}) = \mathbf{P}(t_k) + \int_{t_k}^{t_{k+1}} (\mathbf{P}(\boldsymbol{\tau})\mathbf{A} + \mathbf{A}^T\mathbf{P}(\boldsymbol{\tau}) + \mathbf{Q}) d\boldsymbol{\tau}$
Step 3: Measurement update at sample t_{k+1}
$\tilde{\mathbf{z}}(t_{k+1}) = \mathbf{h}(t_{k+1}, \tilde{\mathbf{x}}(t_{k+1}))$
$\mathbf{H} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}(t_{k+1})} \Big _{[\mathbf{t}_{k+1}, \tilde{\mathbf{x}}(t_{k+1})]}$
$\boldsymbol{\nu}(t_{k+1}) = \mathbf{z}(t_{k+1}) - \tilde{\mathbf{z}}(t_{k+1})$
$\mathbf{S}(t_{k+1}) = \mathbf{H}\tilde{\mathbf{P}}(t_{k+1})\mathbf{H}^T + \mathbf{R}(t_{k+1})$
$\mathbf{W}(t_{k+1}) = \tilde{\mathbf{P}}(t_{k+1})\mathbf{H}^T\mathbf{S}^{-1}(t_{k+1})$
$\hat{\mathbf{x}}(t_{k+1}) = \tilde{\mathbf{x}}(t_{k+1}) + \mathbf{W}(t_{k+1})\boldsymbol{\nu}(t_{k+1})$
$\mathbf{P}(t_{k+1}) = \tilde{\mathbf{P}}(t_{k+1}) - \mathbf{W}(t_{k+1})\mathbf{S}(t_{k+1})\mathbf{W}^T(t_{k+1})$
Step 4: Replace t_k with t_{k+1} , and return to Step 2

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